1(b) Practical: Create a PWA that displays a custom offline page when the Internet is unavailable. (**SHOWS OFFLINE PAGE**)

Student Record Management PWA

1. Create the Project

In Visual Studio 2026: Create a new project → ASP.NET Core Web App (Razor Pages) → .NET 10 → Create.

Project name:

```
StudentPWA
```

2. Create the Student Record Page

Open Pages/Index.cshtml and replace its contents with:

```
@page
@model IndexModel

<div class="container mt-5">

    <h1 class="text-center text-primary">
        Student Record Management
    </h1>

    <p class="text-center">
        B.Sc. Information Technology
    </p>

    <form class="mt-4">

        <div class="mb-3">
            <label>Roll No.</label>
            <input type="text" class="form-control" placeholder="Enter Roll No.">
        </div>

        <div class="mb-3">
            <label>Student Name</label>
            <input type="text" class="form-control" placeholder="Enter Student Name">
        </div>

        <div class="mb-3">
            <label>Class</label>
            <input type="text" class="form-control" placeholder="Enter Class">
        </div>

        <div class="mb-3">
            <label>Email</label>
            <input type="email" class="form-control" placeholder="Enter Email">
        </div>

        <div class="mb-3">
            <label>Mobile No.</label>
            <input type="text" class="form-control" placeholder="Enter Mobile Number">
        </div>

        <button type="button" class="btn btn-primary">
            Add Student
        </button>
```

```
        </form>

    </div>
```

3. Create the PWA Manifest

Inside wwwroot, create manifest.json and add:

```
{
  "name": "Student Record Management PWA",
  "short_name": "StudentPWA",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd"
}
```

The manifest tells the browser that the website is a Progressive Web App.

4. Create the Offline Page

Inside wwwroot, create offline.html and add:

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Offline</title>

    <style>
        body {
            font-family: Arial;
            text-align: center;
            margin-top: 100px;
            background: #f5f7fa;
        }

        .box {
            background: white;
            padding: 40px;
            margin: auto;
            max-width: 450px;
            border-radius: 15px;
            box-shadow: 0 5px 20px #ccc;
        }

        h1 { color: #0d6efd; }

        button {
            padding: 12px 25px;
            background: #0d6efd;
            color: white;
            border: none;
            border-radius: 6px;
        }
    </style>
</head>

<body>
    <div class="box">
        <h1> You Are Offline</h1>
        <p>Internet connection is unavailable.</p>
        <p>Please check your Internet connection and try again.</p>
```

```
        <button onclick="location.reload()">Try Again</button>
    </div>
</body>
</html>
```

5. Create the Service Worker

Inside wwwroot, create sw.js and add:

```
const CACHE_NAME = "student-pwa-v1";
const OFFLINE_PAGE = "/offline.html";

// INSTALL
self.addEventListener("install", event => {

    event.waitUntil(

        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.add(OFFLINE_PAGE);
            })
    );
    self.skipWaiting();
});

// ACTIVATE
self.addEventListener("activate", event => {

    event.waitUntil(
        caches.keys().then(cacheNames => {
            return Promise.all(
                cacheNames
                    .filter(name => name !== CACHE_NAME)
                    .map(name => caches.delete(name))
            );
        })
    );

    self.clients.claim();
});


// FETCH
self.addEventListener("fetch", event => {
    // Handle webpage requests
    if (event.request.mode === "navigate") {
        event.respondWith(
            fetch(event.request)
                .catch(() => {
                    return caches.match(OFFLINE_PAGE);
                })
        );
        return;
    }


    // Handle other resources
    event.respondWith(
        fetch(event.request)
            .catch(() => {
                return caches.match(event.request);
```

```
            })
    );

});
```

6. Register the Service Worker

Open Pages/Shared/_Layout.cshtml. Inside <head>, add:

```
<link rel="manifest" href="/manifest.json">
```

Before the closing </body> tag, add:

```
<script>
    if ("serviceWorker" in navigator) {
        window.addEventListener("load", function () {
            navigator.serviceWorker
                .register("/sw.js")
                .then(function (registration) {
                    console.log(
                        "Service Worker registered:",
                        registration.scope
                    );

                })
                .catch(function (error) {
                    console.log(
                        "Service Worker registration failed:",
                        error
                    );
                });
        });
    }

</script>
```

7. Run the Application

Press Ctrl + F5. The application will open at https://localhost:xxxx/. The Student Record Management page will be displayed.

Test the application as you have already created the **offline page**, then to test whether the PWA displays it when there is no internet:

1. Press **F12 → Application**.
2. Go to **Service Workers**.
3. Check **Offline**.
4. Make sure your **Service Worker is registered and activated**.
5. Go to **Network** → check **Offline** as well.
6. Refresh the page (**Ctrl + R**).